

CMSC 35401-2: Active Representation Learning: Project Guideline

Project Guideline

Students are required to conduct and submit a research project as well as complete a project presentation in an assigned lecture slot during Week 9. Students may work on projects in groups of up to 3 people. Note the topic of your project can be different from the topic of your presentation.

The project can be on any topic related to the course material. The purpose of the project is to gain a better understanding of experimental and/or theoretical aspects of active representation learning algorithms. While we encourage you to explore new research directions, projects may also focus on reproducing or comparing previous approaches, perhaps investigating various theoretical or design choices. If you are a graduate student currently working on a related topic, you may use your research for the project. Generally, projects will fall into one of the following categories:

- **Pure Theory:** study proof techniques, try to extend a proof, or apply a proof to a new setting.
- **Algorithms & Models:** extend an algorithm or model, design a new one, or adapt one for a new setting.
- **Applications:** identify the correct assumptions for approaching a new setting, experimentally validate these assumptions.

Project Report

The final project report should resemble a (mini-) research paper in scope and tone to the papers discussed in class. You are not expected to produce state-of-the-art results or exhaustively explore every aspect of the problem. However, your report should reflect thoughtful engagement with the topic—in how you motivated the problem, designed your approach, and evaluated your results.

The final project report is due by **May 29, 2026** (submission via Gradescope).

Evaluation Criteria

There is no formal grading rubric. Your report will be evaluated based on the depth of thought in three areas:

- **Problem setting:** We will look for a clear motivation of an interesting or challenging problem, and engagement with prior work and its limitations.
- **Approach:** We expect a thoughtful approach that addresses limitations of prior work, along with careful reflection on any implementation challenges.

- **Investigation:** We expect meaningful evaluation metrics, comparisons to relevant baselines, and insightful analysis or visualization of results. For theoretical projects, you should also clearly state assumptions and limitations.

Structure Guide for Project Presentation and Report

You can structure the report however you see fit. We recommend the following structure:

- **Introduction:** Motivate why this area is interesting, describe key technical challenges, and give a short summary of how you have addressed these challenges.
- **Background and previous work:** Briefly review any necessary background material. Give an overview of previous approaches, making sure to highlight their assumptions or limitations. This should not be a laundry list of previous papers.
- **Research question and approach:** If not already described in the previous sections, outline the research question(s) that you are addressing. Describe the approach(es) that you considered, providing justification for your choice. If necessary, you may use this section to introduce any proofs or derivations, which will be included in the appendices.
- **Experiments and results:** If there is an empirical component to your project, describe the experimental set-up in this section. You may include additional details in the appendices if necessary. Make sure to describe the data, models, training scheme, performance metrics, etc. Describe the experimental results, referencing relevant figures and tables.
- **Discussion and conclusion:** Restate the research question, approach, and main findings. Describe any interesting implications of your project, outlining possible directions for future research.

Formatting Guidelines

There is no strict page limit. Most reports are approximately 5–8 pages, with additional pages permitted for references, appendices, figures, etc. We do not have strict formatting requirements. You may use the [NeurIPS LaTeX format](#) (with the “preprint” option).